Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Artificial Intelligence and Machine Learning

Date: 16.12.2025	Test -II	Max. Marks: 50+10
Semester: V	UG	Duration: $1\frac{1}{2}$ Hrs + ½ Hr
Course Title: ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING		Course Code: AI253IA

Part A

Sl No	Questions	M	BT	CO
1.	Examine the effectiveness of the ReLU layer compared to other activation functions (sigmoid and Tanh) in CNNs.	2	4	1
2.	Given the values for input gate (I_t=0.8), forget gate (f_t=0.6), output gate (O_t=0.7), previous cell state (C_{t-1}=0.5), and candidate cell state (C_t=4), calculate the updated cell state C_t using the LSTM cell state update equation	2	2	1
3.	Given x_t=2.5, h_{t-1}=0.9, W_{xh}=0.4, W_{hh}=0.8, and b_h=0.5 calculate the hidden state h_t at time step t using the RNN update equation	2	2	1
4.	List any two differences between LSTM and GRU	2	1	1
5.	Perform max pooling on the given 4×4 feature map using a 2×2 window size with stride 2. Show the resulting pooled output.	2	3	1

3	1	1	3
2	5	0	2
1	4	2	1
4	7	2	4

Part B

Sl No	Questions	M	BT	CO
2a.	How is backpropagation implemented through **matrix multiplication** in Convolutional Neural Networks (CNNs)?	5	3	1
2b.	Assume you are tasked with classifying medical images for the detection of lung cancer, but the dataset consists of a limited number of images. Identify and justify the technique you would use to increase the training set and explain the process in detail.	5	4	1
3a.	Provide a detailed explanation of the mathematical formulations for Bidirectional and Multilayer Recurrent Neural Networks (RNNs), including the state transition equations for both forward and backward passes in BRNNs.	5	3	3
3b	Provide a comprehensive technical overview of a Convolutional Neural Network (CNN) architecture. Please include a brief explanation of each layer, such as the input layer, convolutional layers, activation functions, pooling layers, fully connected layers, and the output layer.	5	2	1
4a.	In a smart grid energy management system, an Echo State Network (ESN) is used to forecast energy demand based on real-time data from IoT sensors. The system receives input features such as current energy consumption, weather conditions, and time of day. The ESN is trained to predict energy demand for the next hour. **a)** Describe the architecture of the ESN in this scenario, highlighting the roles of the input, reservoir, and output layers.	5	3	3
4b.	Consider the statement, *"Head Coach GG fail in Home Test Series."* Discuss briefly how a Long Short-Term Memory (LSTM) network can be effectively utilized for grammatical error detection and correction in such sentences.	5	3	1

	Provide the architecture diagram of an LSTM cell and provide mathematical equations for the cell state			
5a.	Given the 6×6 input matrix and the 3×3 filter shown below, perform the following operations. Use **zero padding** and **stride = 2** for the convolution, treat the filter as a cross-correlation (no kernel flip), and show intermediate 3×3 blocks and calculations where helpful. Input 6*6 matrix 3 0 1 2 7 4 1 5 8 9 3 1 2 7 2 5 1 3 0 1 3 1 7 8 4 2 1 6 2 8 2 4 5 2 3 9 Filter 3*3 1 0 -1 1 0 -1 1 0 -1 $\frac{6-3+2}{2}+1$ a. Compute the 2D convolution (cross-correlation) output with zero padding, **stride = 2**. Show the 3×3 blocks used and the resulting output matrix. b. Apply **ReLU activation** to the convolution output and show the result. c. Perform **max pooling** on the ReLU output using a **2×2 window** with **stride = 2**. Show the pooling windows and the final pooled output.	6	3	2
5b.	Analyze the different topologies of Recurrent Neural Networks (RNNs) and discuss how each topology is applied in real-world scenarios.	4	4	3
6.	Consider the convolutional neural network defined by the layers in the left column below. Fill in the shape of the output volume and the number of parameters at each layer. You can write the shapes in the NumPy format (e.g. (128,128,3)) (Perform step by step calculations) Notation: a. CONV5-N denotes a convolutional layer with N filters with height and width equal to 5. Padding is 2, and stride is 1. b. POOL2 denotes a 2x2 max-pooling layer with stride of 2 and 0 padding. c. FC-N denotes a fully connected layer with N neurons.	10	3	2

Layer	Activation Volume Dimensions	Number of parameters
Input	32*32*1	0
CONV5-10		
POOL2		
CONV5-10		
POOL2		
FC10		